

MST-Technologies and Processes

March, 24th, 2023

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points: **50 of 100 in total**

Points: _____

Result: _____



1. Processes

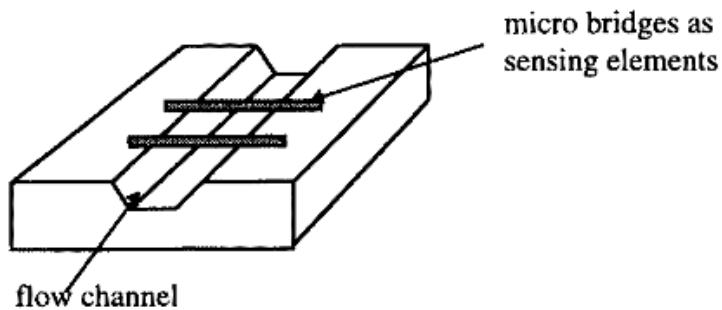
27

1.1 Manufacturing of a microfluidic sensor

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Sketch the manufacturing steps required to fabricate the following structure which is a microfluidic channel with an anemometric sensor spanned across it. The substrate is silicon (you are free to choose the cut), the micro bridges are made from gold.

Sketch and briefly explain each step. Make sure to include all the relevant labels.



1.2 We have three different categories of processes in MST, name them and sort the following processes into the right category:

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- Metallization
- Development during lithography
- Thermal oxidation



2. Thermal oxidation of silicon

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2.1 In the Deal-Grove model, the following variables are used:

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- D as the diffusivity of the oxidant in silicon dioxide
- k as the reaction rate of the oxidation of silicon to silicon dioxide
- N as the number of oxidant molecules required to produce one unit volume of oxide
- C as the ambient concentration of the oxidant
- $A = \frac{2D}{k}$ as a simplification factor
- $B = \frac{2DC}{N}$ as a second simplification factor

State the equation for the layer thickness $x(t)$ at lower layer thicknesses, i.e., for $x(t < t_c)$.

2.2 For usage in the Deal-Grove model we determined the following factors:

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- $A = 0.4 \mu\text{m}$
- $B = 0.04 \mu\text{m}^2/\text{h}$

What is the characteristic time t_c ? *[You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]*

2.3 Using the same parameters as above, we now run an oxidation process for 100 hours (which is longer than t_c). What is the expected layer thickness we achieve? *[You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]*

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3. Cleanroom

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- 3.1 Draw a cross-section of a typical cleanroom. Label the relevant components. Make sure to include (at least) the following items: 8
- grey area
 - white area
 - plenum
 - bypacks
 - media supply
- 3.2 There are three types of cleanroom setups which are commonly used: 9
- name them
 - give one advantage for each
 - give one disadvantage for each
- 3.3 Standard filter-fan units (FFUs) use four filtration effects. Name them and shortly explain each of them. 6



4. Semiconductor materials

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- | | | |
|-----|---|---|
| 4.1 | Name two common semiconductor materials. | 2 |
| 4.2 | What is their most outstanding physical properties (for which they would be most likely chosen in an application)? | 2 |
| 4.3 | Sketch and explain the <u>Czochralski</u> process for manufacturing <u>single-crystalline</u> silicon. Make sure to label each component in your drawing. | 8 |

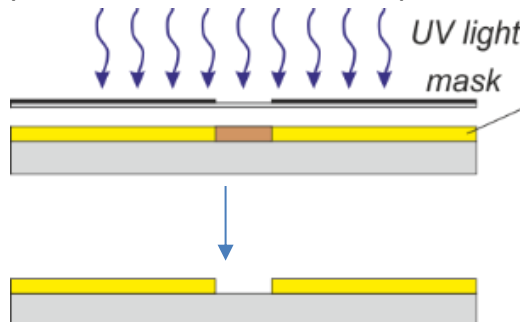


5. Photolithography

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- 5.1 The following image depicts two steps in a lithography workflow: the exposure (top) and the development step (below). Which type of photoresist was used here (what should the missing label read)?

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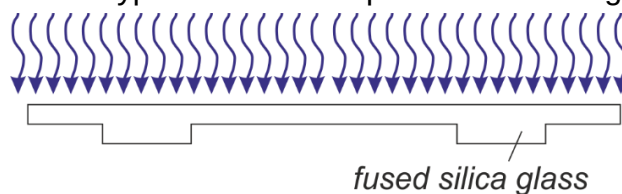


- 5.2 What would be a typical photoresist (brand name, photochemistry on which it is based, etc.) used for the setup depicted in 5.1?
- 5.3 What would you use as developer solution for the photoresist in the setup depicted in 5.1 and the type of photoresist you chose in 5.2?
- 5.4 Which type of mask is depicted in this image?

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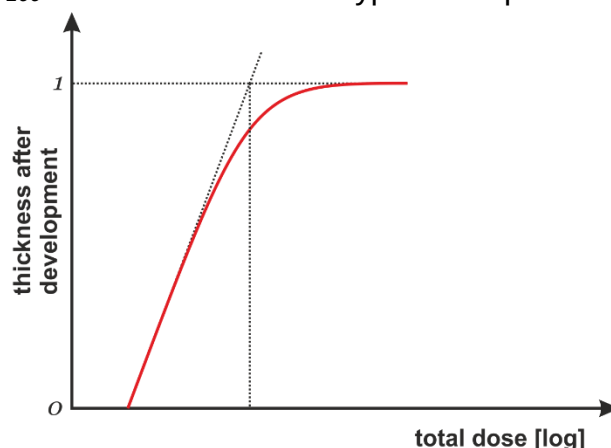


- 5.5 What are the two parameters that a mask can modulate on the incoming light?
- 5.6 Sketch the lithographic setup in proximity lithography. Make sure to include all relevant labels in your sketch.
- 5.7 During characterization of an old bottle of photoresist via an exposure study, you obtain the following graph. Label the characteristic D_0 and D_{100} values and state the type of the photoresist.

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- 5.8 When choosing the substrate material of a photomask you have several options. What are the most common types of glasses and which one

4



would you chose if you plan to run a lithography at a wavelength of 200 nm?

- 5.9 Sketch the classical lithography workflow. Make sure to include all relevant labels. 4

